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Interactions of the organogold(III) compound Aubipyc with the copper chaperone Atox1: a joint mass spectrometry and circular dichroism investigation

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Abstract The so called "copper trafficking system" in mammalian cells is primarily devoted to the regulation of copper transport and homeostasis. This system, now well characterized, consists of a few strictly interconnected proteins that assist copper entrance inside cells and then promote metal transfer and delivery to essential copper-dependent cellular proteins (Boal and Rosenzweig 2009a; Banci et al., Mol Life Sci 67:2563–2589, 2010). Yet, the "copper trafficking system" may also facilitate the entrance inside cells of non-physiological metal species such as clinically established platinum drugs. ESI and MALDI MS methods are exploited here to characterize the interactions occurring between the experimental anticancer organogold(III) drug, Aubipyc, and the copper chaperone Atox1, a key protein of the copper trafficking system. The nature of the adducts that are formed when reacting Aubipyc with Atox1 is elucidated in detail. Characterization of the Aubipyc/Atox1 system is further supported by circular dichroism experiments. Binding competitions with mercury and bismuth ions were also explored. The relevance and the biological implications of the present results are discussed.

Keywords Anticancer drugs · Gold compounds · ESI-MS · Circular dichroism · Protein melting · Protein metalation

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Introduction

Atox1 is an intracellular metallochaperone protein crucially involved in copper trafficking; details of Atox1 structure and function were extensively clarified through a few recent studies (Rosenzweig and O' Halloran 2000; Hussain and Wittung-Stafshede 2007). Most Atox1 homologues are 70 amino acid proteins containing a conserved CXXC motif for copper(I) binding in the vicinity of the N-terminus. Detailed structural information on Atox1 and its copper complex is now available (Rosenzweig et al. 1999). The copper(I) center is bound to two cysteines i.e. Cys 15 and Cys 12. The nature of the copper(I) ligands in the metal binding site of Atox1 renders this site well prone to accommodate, beyond copper, a variety of "soft" metal ions and metallodrugs. Accordingly, it was suggested and experimentally proved that the copper trafficking system may play a key role in the cellular uptake and processing of platinum-based anticancer drugs with relevant pharmacological consequences. A few studies documented that cisplatin reacts with Atox1 forming stable derivatives (Boal and Rosenzweig 2009a; Safaei and Howell 2005; Howell et al. 2010; Katano et al. 2002). Structures of cisplatin derivatives of Atox1 were then solved through X-ray diffraction (Boal and Rosenzweig 2009b) or NMR measurements (Arnesano et al. 2011) showing a direct interaction of the platinum(II) center with the CXXC motif. In turn, a study by Pernilla Wittung et al. investigated the reaction of cisplatin with Atox1 through a variety of biophysical methods and demonstrated that copper(I) and platinum(II) ions may bind simultaneously to this protein to nearby, yet independent and not mutually exclusive, anchoring sites (Palm et al. 2011; Palm-Espling and Wittung-Stafshede 2012). Very recently we have demonstrated tight binding to Atox1 of a variety of gold compounds of medicinal interest (Gabbiani et al. 2012).

Mass spectrometry methods constitute a very powerful and straightforward tool to analyze, at the molecular level, the metal binding properties of Atox1 toward different metal species. The suitability of ESI MS to characterize the binding of various gold species to Atox1 was recently highlighted (Gabbiani et al. 2012); here, we further exploit this approach to elucidate, in deeper detail, the nature of the gold binding site in the Aubipyc/Atox1 adduct. Conversely, MALDI studies of tryptic digests of the Aubipyc/Atox1 derivative allow identification of the protein portion containing the gold binding site. In addition, variable-temperature circular dichroism (CD) experiments were carried out to determine the changes of Atox1 melting temperature profile induced by gold binding. Subsequently, the metal binding properties of Atox1 toward a group of thiophilic metal species were explored, particular attention being paid to a few exogenous metals of medicinal or toxicological relevance such as bismuth and mercury. The possible competition for the same protein binding site between the gold drug Aubipyc and the above metal species was eventually investigated. A more comprehensive picture of the metal binding features of Atox1 protein emerges from comparative analysis of the various results here reported.

Materials and methods

Materials

Aubipyc was synthesized as previously described (Marcon et al. 2002); $SbCl_3$, As_2O_3, $Bi(NO_3)_3$ were purchased from Sigma Aldrich, $HgCl_2$ was purchased from Merck and used without further purification. Copper transport protein Atox1 was purchased from Giotto Biotech: MW = 7.4 kDa. Full length HAH1 (Copper transport protein Atox1, Metal transport protein Atox1) cloned from human cDNA, expressed in *E. Coli*; purity >95 % by SDS-PAGE. Atox1/Atox1-M mixture where M = Methionine (due to *E. Coli*) HgCl2.

CD experiments

CD measurements were run on a solution of Atox1 50 μM in NaCl 50 mM, pH 7 (apo form), additioned with DTT 1 mM and 1 equiv of Aubipyc 50 μM (holo form), prepared directly in a 1-cm quartz cylindrical cell (2.5 ml). The holo form was incubated at room temperature (27 °C) for 12 h before measurement. CD spectra were measured with a Jasco J-715 spectropolarimeter with the following conditions: scan speed 50 nm/min; response 1 s; data pitch 0.1 nm; bandwidth 2.0 nm; 4 accumulations. Variable-temperature spectra were run with a home-made Peltier apparatus purposely designed for 1-cm cylindrical cell, equipped with an Ascon MS 30 controller (±1 °C accuracy). CD data were acquired at 220 nm with a scan rate of 1°/min and the following conditions: response 2 s; data pitch 1 s; bandwidth 2.0 nm. The recovery of the CD signal after cooling back at 30 °C was above 80 % in all cases (see SI). The statistical treatment of data is discussed in the Supporting Information.

Digestion and MALDI-TOF analyses

Two 100 μM solutions of Atox1/Atox1-M in ammonium acetate buffer solution at pH 6.8 containing DTT (500 μM) were prepared with or without adding Aubipyc (protein to metal ratio 1:1, at RT for 24 h,

N_2 inert atmosphere). To 5 µL of both solutions were added 30 µL of Ambic 40 mM. Both solutions were digested on immobilized trypsin tips (DigesTip, ProteoGen Bio, Siena, Italy) and 0.4 µL of tryptic digests were spotted on an AnchorChip 400 MALDI target (Bruker Daltonics, Bremen, Germany) together with 0.4 µL of matrix solution (10 mg/ml α-cyano-4-hydroxycinnamic acid in 70/30 acetonitrile/0.1 % TFA). The solvent was allowed to evaporate and the two dried spots were analyzed with a MALDI-TOF spectrometer (Ultraflex III MALDI-TOF/TOF, Bruker Daltonics, Bremen, Germany). The working conditions were the following: reflectron positive mode, mass range 850–5000 m/z. The spectra were processed using Flex Analysis software 3.0 from Bruker Daltonics.

ESI-MS analyses

For the analysis Atox-1 was dissolved (10^{-4} M) in 20 mM ammonium acetate buffer (AmAc), pH 6.8 containing DTT (1:5 protein:DTT molar ratio). Then the salts or/and complex were added (1:1 or 5:1 metal/protein ratio) to the solution and incubated at room temperature for 24 h in N_2 inert atmosphere. Samples were diluted with 1 % HCOOH to a final concentration of 10 µM and immediately analysed by direct introduction (5 µL/min) into an Orbitrap high-resolution mass spectrometer (Thermo, San Jose, CA) equipped with a conventional ESI source. The working conditions were the following: spray voltage 3.1 kV, capillary voltage 45 V, capillary temperature 220 °C and tube lens voltage 230 V. Sheath gas and auxiliary gas were set, respectively, at 17 (arbitrary units) and 1 (arbitrary units). For acquisition, Xcalibur 2.0. software (Thermo) was used and monoisotopic and average deconvoluted masses were obtained by using the integrated Xtract tool. For spectra acquisition a nominal resolution (at m/z 400) of 100,000 was used.

Results and discussion

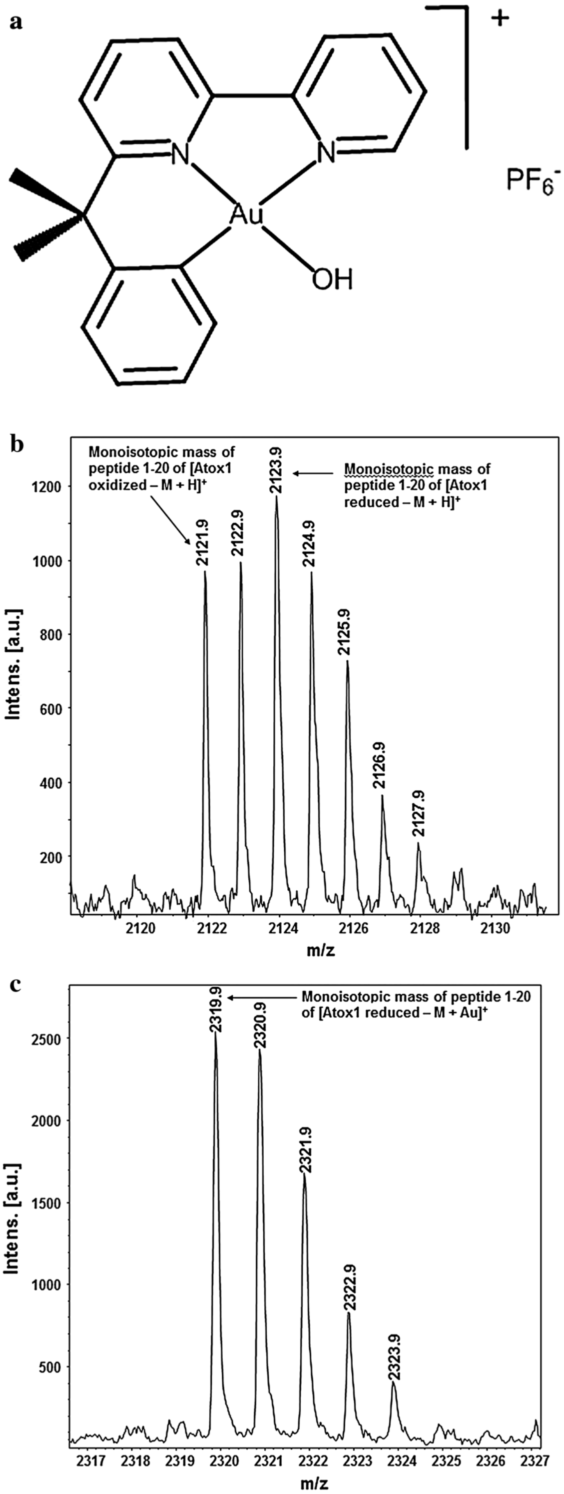
ESI and MALDI MS characterization of gold binding in Aubipic/Atox1

It was previously shown that Atox1 may form stable derivatives with a variety of gold compounds of medicinal interest. In particular, Aubipyc (Fig. 1a) is an established organogold(III) complex with promising antiproliferative and anticancer properties, extensively studied in recent years (Nobili et al. 2010). Notably, under the solution conditions used to monitor metal binding to Atox1, i.e. within the strongly reducing environment created by the addition of DTT, the gold(III) center of Aubipyc is reduced to gold(I), the terdentate ligand is released and the resulting gold(I) species binds Atox1. The main derivative that is formed corresponds to a single bare gold(I) ion bound to Atox1 (Gabbiani et al. 2012). We decided to study in more depth this derivative and identify the exact location of the gold(I) binding site. Adduct formation is well witnessed by deconvoluted ESI MS spectra showing a clear peak at ~7465 Da, in agreement with previously published work (Gabbiani et al. 2012).

Based on simple coordination chemistry considerations and HSAB concepts, it is straightforward to hypothesize that gold(I) binding may occur at the level of the copper(I) binding site. However, to better ascertain this point, tryptic digestion was carried out either on the Atox1/Atox1-M mixture alone or on the same mixture incubated with Aubipyc. Then, MALDI spectra were recorded on both tryptic digests. Notably, the most abundant tryptic fragment obtained from the first mixture was the 1–20 peptide of Atox1-M (Fig. 1b). Upon comparing this spectrum with the one obtained from tryptic digestion of the second mixture (i.e. Atox1/Atox1-M incubated with Aubipyc) we noticed the appearance of a relatively intense peak at 2319.8 Da due to the binding of Au+ to 1–20 peptide of reduced Atox1-M (Fig. 1c). Further, the absence of peaks at 2317 Da implies that the oxidized form of peptide 1-20 is unable to coordinate the Au^+ ion. This result is nicely consistent with the hypothesis that gold binding takes place at the CXXC motif, as reported for cisplatin by Boal and Rosenzweig (2009a).

Stability of the Aubipyc/Atox1 adduct assessed by circular dichroism experiments

Circular dichroism (CD) is a well-known technique to characterize protein thermal stability and follow the thermodynamics of unfolding processes (Greenfield 2006). Previous CD studies (Rosenzweig and O' Halloran 2000; Palm et al. 2011) on Atox1 showed that the copper-bound holo form is only slightly more



◀**Fig. 1** **a** Structure of Aubipyc. **b** MALDI spectrum of 1–20 fragment obtained through tryptic digestion of Atox1-M. **c** MALDI spectrum of 1–20 fragment obtained through tryptic digestion of Atox1-M + Aubipyc (incubated for 24 h at RT, 1:1 complex to protein ratio in 20 mM ammonium acetate buffer solution, N_2 inert atmosphere)

resistant toward thermal unfolding than the apo form (melting points of 71 ± 1 and 68 ± 1 °C, respectively). After incubation with Aubipyc, the CD spectrum of Atox1 shows just a small intensity decrease (~10 %) but retains the same shape (Fig. 2a), indicating that the protein does not experience any appreciable change in its secondary structure. The CD spectrum of the Aubipyc/Atox1 derivative is stable over a period of 24 h. Melting experiments were comparatively run in the 30–95 °C interval on Atox1 and Aubipyc/Atox1 samples and followed by means of the CD signal at 220 nm, where the maximum CD amplitude occurs. The two melting curves demonstrate immediately a strong difference between the two species (Fig. 2b), in the sense that even at the highest accessible temperature the Aubipyc/Atox1 derivative is far from complete unfolding. A non-linear least-square fitting of the two traces with a classical two-state unfolding model (Greenfield 2006; Kemmer and Keller 2010) (see details in the SI) affords melting points of 72 ± 1 °C for Atox1 and 93 ± 3 °C for the Aubipyc/Atox1 derivative. Although the accuracy of this latter value is diminished by the incomplete melting process, even admitting the largest possible experimental error, gold(I) binding seems to stabilize the protein by at least 15 °C toward thermal unfolding. This value is especially noteworthy in comparison with the much more limited effect observed for copper.

Metal binding properties of Atox1 versus other “soft” metals monitored through ESI MS

Next, an ESI MS approach (analogous to that reported for medicinal gold compounds in our previous work) (Gabbiani et al. 2012) was exploited to assess Atox1 metal binding abilities toward a variety of metal species of medicinal or toxicological interest (Dabrowiak 2009; Salvador et al. 2012) and to perform competition studies with Aubipyc. For this purpose, we selected a number of metal (or metalloid) species, namely antimony, arsenic, bismuth and mercury, exhibiting a pronounced thiophilic character. At first, a screening was carried out for each metal under the same experimental conditions. As the result of this

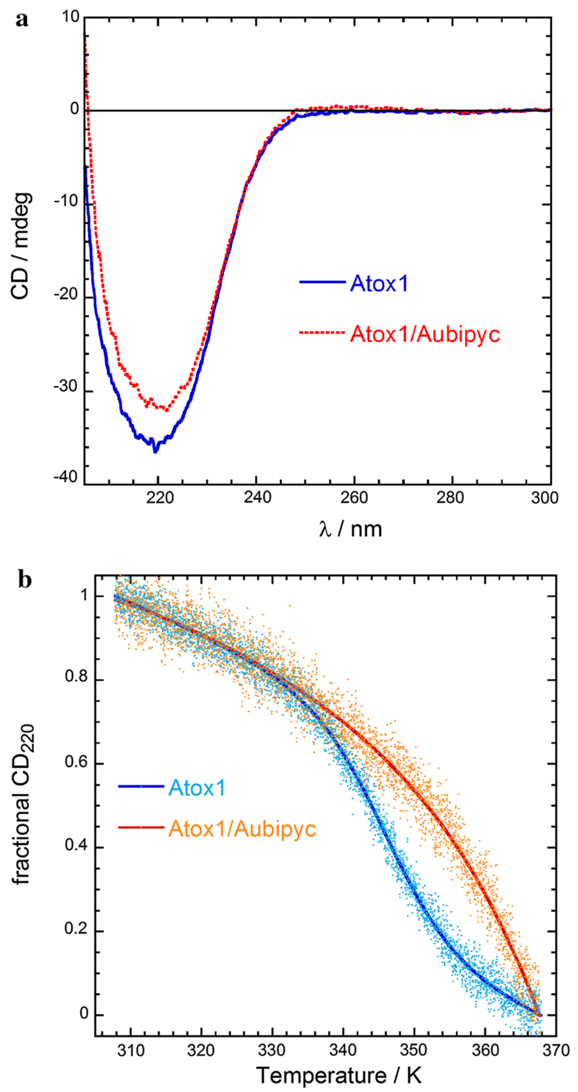


Fig. 2 **a** CD spectra of Atox1 and its 1:1 derivative with Aubipyc, 50 µM in NaCl 50 mM pH 7, DTT 1 mM, 28 °C, 1 cm cell. **b** Thermal unfolding of Atox1 and its 1:1 derivative with Aubipyc, same conditions as above, monitored by CD at 220 nm. The fitting curves are adapted from the least-square fitting of raw data shown in the Supporting Information

preliminary screening, adduct formation was demonstrated for bismuth and mercury while the other species failed to form adducts, at least under the here adopted experimental conditions. Deconvoluted ESI MS spectra documenting adduct formation in the case of mercury and bismuth are shown in Fig. 3. Notably, the peak of the adduct, in the case of mercury, is significantly higher than that of the native apoprotein implying a large binding affinity of mercury for Atox1.

For both species, i.e. mercury and bismuth ions, the ESI-MS spectrum is dominated by a major peak corresponding to the monometalated derivative, suggesting the existence of a single metal binding site. In the case of bismuth we could ascertain that the abundance of this derivative increases with increasing the metal: protein ratio, with no evidence of bismetalated species (see SI).

The structure of a Hg-Atox1 derivative was previously reported by Boal and Rosenzweig (2009a) within a comparative crystallographic study; in contrast, formation of a tight adduct between Atox1 and

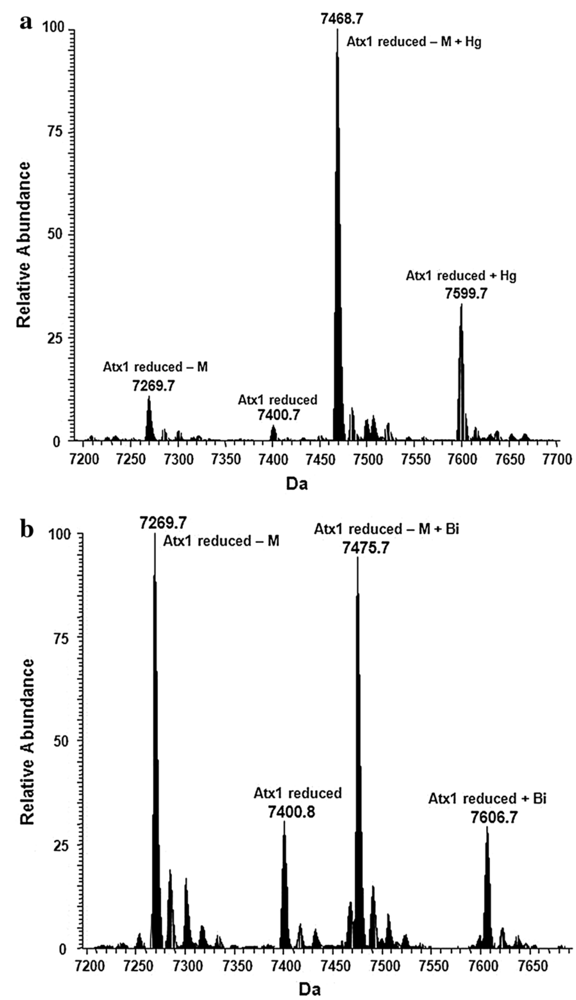


Fig. 3 Deconvoluted ESI MS spectra of mixture Atox1/Atox1-M incubated with **a** $HgCl_2$ and **b** $Bi(NO_3)_3$ for 24 h at RT. Experiments at 1:1 metal to protein ratio in 20 mM ammonium acetate buffer solution, N_2 inert atmosphere

the tripositive bismuth(III) ion is described here for the first time.

Metal competition studies

Having documented the strong affinity of the above metal species—i.e. Aubipyc, bismuth(III) ions and mercury (II) ions—for Atox1, we moved to investigate possible competition among them in protein binding. In principle, the observation of a direct competition might support the hypothesis that these metal species share the same metal binding site; in addition initial information may be inferred on the relative stabilities of the various metal-protein species. This goal could be accomplished straightforwardly through a single experiment where Atox1 was challenged simultaneously against bismuth, mercury and gold, presented at 1:1 metal to protein ratio, under the same experimental conditions reported above. The resulting sample was analyzed through ESI-MS and found to exhibit a single major peak at 7468 Da, corresponding to the Hg-Atox1 derivative (Fig. 4).

Remarkably, no evidence was obtained for any Bi-Atox1 or Au-Atox1 derivatives. These spectral features strongly suggested that the three tested metal species, namely Hg, Bi and Aubipyc do compete for the same metal binding site or region which realistically corresponds to the CXXC motif. Mercury turns out to be- by far- the strongest agonist for the Atox1 metal binding site, completely abolishing the binding of either bismuth or Aubipyc to the same site. Strong evidence of a direct competition between the three considered metal species is thus offered.

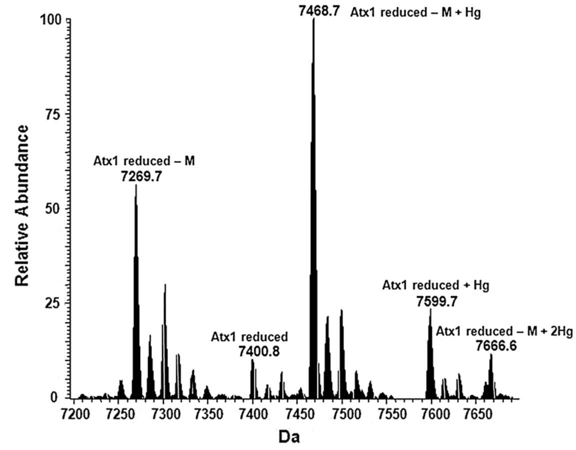


Fig. 4 Deconvoluted ESI-MS spectrum of competition experiment between Au, Hg, Bi incubated with Atox1/Atox1-M mixture for 24 h at RT, 1:1 metals to protein ratio in 20 mM ammonium acetate buffer solution, N_2 inert atmosphere

Conclusions

Overall, with the present investigation, we succeeded in elucidating some key aspects of the reaction of the organogold(III) species Aubipyc with Atox1 protein and its competition with other metal species. Though being primarily deputed to copper handling and processing, Atox1 has revealed prominent binding properties even toward other metal species in relation to its pronounced thiophilic character; for instance, extensive protein binding of several platinum and gold compounds of medicinal interest was recently demonstrated with important biological and pharmacological implications.

Here, we have first deepened the characterization of the Atox1 binding properties toward the cytotoxic organogold(III) complex Aubipyc. We have shown that Aubipyc, in its reaction with Atox1, mainly produces a monometalated gold(I)/Atox1 derivative; that gold binding stabilizes by >15 °C the protein toward thermal unfolding; that gold(I) binding primarily occurs at the tryptic peptide 1–20 in its reduced form; that gold(I) binding is completely inhibited by mercury(II) chloride. Direct competition experiments with mercury strongly suggest that gold(I) binding occurs at the primary metal binding site of Atox1 i.e. the CXXC copper(I) binding site. Also, the formation of a tight adduct between Atox1 and bismuth(III) has been demonstrated, here, for the first time through ESI MS experiments. More in general, the formation of stable adducts between Atox1 and various metal species (mostly thiophilic metals), that are straightforwardly documented here through MS experiments, suggests that this protein of the copper trafficking system may play a major role in the intracellular processing of numerous non-physiological metal compounds of pharmacological and/or toxicological relevance.

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